Ex.No:5 **Date: 1/9/25**

SIMPLE CALCULATOR USING XMLRPC

Aim:

To develop a simple calculator using xmlrpc in python.

Introduction: XML-RPC:

XML-RPC is a protocol that uses XML to encode remote procedure calls (RPC) and HTTP as a transport mechanism, allowing a client to call functions on a remote server over a network.

Algorithm:

Client:

1. Import xmlrpc.client and create a ServerProxy for http://localhost:8000/.
2. Loop a fixed number of times (e.g., 5).
3. Prompt the user to input two numbers (a and b).
4. Call server functions (add, sub, mul, div, mod) via the proxy.
5. Receive results from the server for each operation.
6. Print the results to the user.

Server:

1. Import SimpleXMLRPCServer from xmlrpc.server.
2. Define arithmetic functions: add, sub, mul, div, mod.
3. Create an XMLRPCServer object bound to localhost on port 8000.
4. Register each function with the server.
5. Print a message indicating the server is listening.
6. Start the server using serve_forever() to handle incoming client requests.

Program Code:

Client:

```
import xmlrpc.client
proxy= xmlrpc.client.ServerProxy('http://localhost:8000/')
for i in range(5):
    a=int(input("Enter a number:"))
    b=int(input("Enter b number:"))
    print("addition of given number is %d "%((proxy.add(a,b))))
    print("sub of given number is %d "%((proxy.sub(a,b))))
    print("mul of given number is %d "%((proxy.mul(a,b))))
    print("div of given number is %d "%((proxy.div(a,b))))
    print("mod of given number is %d "% ((proxy.mod(a,b))))
```

Server:

```
from xmlrpc.server import SimpleXMLRPCServer
def add(a,b):
    return a+b
def sub(a,b):
    return a-b
def mul(a,b):
    return a*b
def div(a,b):
    return a/b
def mod(a,b):
    return a%b
server = SimpleXMLRPCServer(("localhost", 8000))
print("Listening on port 8000...")
server.register_function(add, "add")
```

```
server.register_function(sub, "sub")
server.register_function(mul, "mul")
server.register_function(div, "div")
server.register_function(mod, "mod")
server.serve_forever()
```

Output:

Client:

```
C:\Users\ponsu\Documents>python client1.py
Enter a number:3
Enter b number:5
addition of given number is 8
sub of given number is -2
mul of given number is 15
div of given number is 0
mod of given number is 3
Enter a number:7
Enter b number:8
addition of given number is 15
sub of given number is -1
mul of given number is 56
div of given number is 0
mod of given number is 7
Enter a number:77
Enter b number:89
addition of given number is 166
sub of given number is -12
mul of given number is 6853
div of given number is 0
mod of given number is 77
Enter a number:65
Enter b number:65
addition of given number is 130
sub of given number is 0
mul of given number is 4225
div of given number is 1
mod of given number is 0
Enter a number:45
Enter b number:90
addition of given number is 135
sub of given number is -45
mul of given number is 4050
div of given number is 0
mod of given number is 45
```

Server:

```
C:\Users\ponsu\Documents>python server1.py
Listening on port 8000...
127.0.0.1 - - [27/Sep/2025 18:12:06] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:12:08] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:12:10] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:12:12] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:12:14] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:12:50] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:12:52] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:12:54] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:12:56] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:13:13] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:13:15] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:13:17] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:13:19] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:13:22] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:13:29] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:13:31] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:13:33] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:13:35] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:13:37] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:13:44] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:13:46] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:13:48] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:13:50] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:13:52] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:14:01] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:14:03] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:14:05] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:14:08] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:14:10] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:14:18] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:14:20] "POST / HTTP/1.1" 200 -
127.0.0.1 - - [27/Sep/2025 18:14:22] "POST / HTTP/1.1" 200 -
```

Result:

Thus a simple calculator was developed using xmlrpc in python.